NAYEONG CHO

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EDUCATION

M.S. in Atmospheric Sciences, Yonsei University, Republic of Korea *2011*
Thesis: Analysis of spectral enhancement of UV and visible radiation due to clouds by Brewer Spectrophotometer observation and Monte Carlo radiative transfer model calculation

B.S. in Atmospheric Sciences, Yonsei University, Republic of Korea *2009*

EMPLOYMENT

Assistant Research Scientist, Earth Science *Dec 2021–present*
UMBC / GESTAR II; NASA Goddard Space Flight Center, Climate and Radiation Branch (Code 613)

Associate Scientist, Earth Science *Oct 2018–Nov 2021*
USRA / GESTAR; NASA Goddard Space Flight Center, Climate and Radiation Branch (Code 613)

Scientist 1 *Jun 2016–Sep 2018*
USRA / GESTAR; NASA Goddard Space Flight Center, Climate and Radiation Branch (Code 613)

Visiting Professional I *Jun 2013–May 2016*
USRA / GESTAR; NASA Goddard Space Flight Center, Climate and Radiation Branch (Code 613)

AWARDS

Best First-Authored Paper, Climate and Radiation Laboratory (613), NASA GSFC *Mar 2026*

Best First-Authored Paper, Climate and Radiation Laboratory (613), NASA GSFC *Feb 2022*

Outstanding Performance Science, Atmospheric Earth Science Labs (610AT), NASA GSFC *Nov 2016*

PROFESSIONAL SERVICE AND EXPERIENCE

Newsletter Editor, Korean Atmospheric Scientist in America *2020.7 – 2021.7*

Newsletter Associate Editor, Korean Atmospheric Scientist in America *2019.8 – 2020.6*

Administrative Managers for Air Monitoring Site, SYU in SKYNET *2009.9 – 2011.8*

Visiting student at UCLA *2010.6 – 2010.7*

Hosting Staff: The 3rd Asia Pacific Radiation Symposium (APRS) *2010.8*

Teaching Assistant, Department of Atmospheric Sciences, Yonsei University, Courses: Atmospheric Chemistry (Fall 2009) *2009.9 – 2010.2*

Teaching Assistant, Department of Atmospheric Sciences, Yonsei University, Courses: Atmospheric Physics (Spring 2010) *2010.3 – 2010.8*

PROFESSIONAL MEMBERSHIPS

American Geophysical Union (AGU), regular member, 2013–present

MEDIA AND PUBLIC ENGAGEMENT

Featured in NASA People: "Cho and Lee, Atmospheric Science and Each Other" (NASA Goddard, 2018).
https://www.nasa.gov/feature/goddard/2018/cho-and-lee-two-loves-atmospheric-science-and-each-other

PEER-REVIEWED PUBLICATIONS

Oreopoulos L., **N. Cho** and D. Lee. (2025). The role of Earth's major cloud systems in the hemispheric albedo symmetry., Journal of Climate, 38 (23): 7203–7215 [10.1175/jcli-d-25-0242.1] [Journal Article/Letter]

N. Cho, L. Oreopoulos, D. Lee, et al. (2025). Describing seasonal mixtures of cloud regimes via "regimes of regimes". Journal of Climate [10.1175/jcli-d-24-0275.1]

Lee D., L. Oreopoulos and **N. Cho**. (2024). Regimes of Cloud Vertical Structure From Active Observations. Journal of Geophysical Research: Atmospheres 130 (1): [10.1029/2024jd041716]

Chen Y., J. Haywood, Y. Wang, …& **Cho, N**… (2024), Substantial cooling effect from aerosol-induced increase in tropical marine cloud cover. Nature Geoscience [10.1038/s41561-024-01427-z]

Oreopoulos, L., **N. Cho**, and D. Lee. (2022), Revisiting cloud overlap with a merged dataset of liquid and ice cloud extinction from CloudSat and CALIPSO. Frontiers in Remote Sensing, 3: [10.3389/frsen.2022.1076471]

Chen, Y., J. Haywood, Y. Wang, ... & **Cho, N**…. (2022), Machine learning reveals climate forcing from aerosols is dominated by increased cloud cover. Nature Geoscience, [10.1038/s41561-022-00991-6]

Oreopoulos, L., **N. Cho**, D. Lee, M. Lebsock, and Z. Zhang. (2022), Assessment of Two Stochastic Cloud Subcolumn Generators Using Observed Fields of Vertically Resolved Cloud Extinction. Journal of Atmospheric and Oceanic Technology, [10.1175/jtech-d-21-0166.1]

Tan, J., **N. Cho**, L. Oreopoulos, and P. Kirstetter (2021), Evaluation of GPROF V05 Precipitation Retrievals under Different Cloud Regimes. Journal of Hydrometeorology, 23 (3): 238-402 [10.1175/jhm-d-21-0154.1]

Jin, D., L. Oreopoulos, D. Lee, J. Tan, and **N. Cho** (2021), Cloud-Precipitation Hybrid Regimes and their Projection onto IMERG Precipitation Data. Journal of Applied Meteorology and Climatology, 60 (6): 733-748 [10.1175/jamc-d-20-0253.1]

Cho, N., J. Tan, and L. Oreopoulos (2021), Classifying planetary cloudiness with an updated set of MODIS Cloud Regimes. Journal of Applied Meteorology and Climatology, [10.1175/jamc-d-20-0247.1]

Lee, D., L. Oreopoulos, and **N. Cho** (2020), An evaluation of clouds and radiation in a large-scale atmospheric model using a cloud vertical structure classification. Geoscientific Model Development, 13 (2): 673-684 [10.5194/gmd-13-673-2020]

Oreopoulos, L., **N. Cho**, and D. Lee (2020), A Global Survey of Apparent Aerosol-Cloud Interaction Signals. Journal of Geophysical Research: Atmospheres, 125 (1): [10.1029/2019jd031287]

Tan, I., L. Oreopoulos, and **N. Cho** (2019), The Role of Thermodynamic Phase Shifts in Cloud Optical Depth Variations With Temperature. Geophysical Research Letters, 46 (8): 4502-4511 [10.1029/2018gl081590]

Jin, D., L. Oreopoulos, D. Lee, **N. Cho**, and J. Tan (2018), Contrasting the co-variability of daytime cloud and precipitation over tropical land and ocean, Atmos. Chem. Phys., 18, 3065-3082, https://doi.org/10.5194/acp-18-3065-2018

Oreopoulos, L, **N. Cho**, and D. Lee (2017b), New insights about cloud vertical structure from CloudSat and CALIPSO observations, J. Geophys. Res. Atmos., 122, 9280–9300, doi:10.1002/2017JD026629.

Oreopoulos, L, **N. Cho**, and D. Lee (2017a), Using MODIS cloud regimes to sort diagnostic signals of aerosolcloud-precipitation interactions, J. Geophys. Res. Atmos., 122, 5416–5440, doi:10.1002/2016JD026120.

Malavelle, F. F., Haywood, J. M., Jones, A., Gettelman, A., Clarisse, L., Bauduin, S., ... & **Cho, N**…. (2017). Strong constraints on aerosol–cloud interactions from volcanic eruptions. Nature, 546(7659), 485-491. doi:10.1038/nature22974

Leinonen, J., M. D. Lebsock, L. Oreopoulos, and **N. Cho** (2016), Interregional differences in MODIS-derived cloud regimes. Journal of Geophysical Research: Atmospheres, 121: [10.1002/2016jd025193]

Oreopoulos, L., **N. Cho**, D. Lee, and S. Kato (2016), Radiative effects of global MODIS cloud regimes, J. Geophys. Res. Atmos., 121, doi:10.1002/2015JD024502.

Oreopoulos, L., **N. Cho, D**. Lee, S. Kato, and G. J. Huffman. (2014), An examination of the nature of global MODIS cloud regimes, Journal of Geophysical Research-Atmospheres, 119 (13): [10.1002/2013JD021409]

Kim, J.*, H. K. Cho, J. Mok, H. D. You, and **N. Cho** (2012), Effects of ozone and aerosol on surface UV radiation variability, Journal of Photochemistry and Photobiology B: Biology, 119, 46-52 http://dx.doi.org/10.1016/j.jphotobiol.2012.11.007

Kim, J.*, S. S. Park, **N. Cho**, W. Kim, and H. K. Cho (2011), Recent variations of UV irradiance at Seoul 2004~2010, Atmosphere, 21(4), 429-438

Park, S. S., J. Kim*, **N. Cho**, Y. G. Lee, and H. K. Cho (2011), The variations of stratospheric ozone over the Korean Peninsula 1985~2009, Atmosphere, 21(4), 349-359

SELECTED CONFERENCE PRESENTATIONS

2026 45th CERES Science Team Meeting – Oct 2026, Title: A Cloud System View of the Changing Planetary Energy Budget

2025 42nd CERES Science Team Meeting – May 2025, Title: A different perspective on the role of clouds in the symmetry of hemispheric reflected solar radiation.

2023 AGU Fall Meeting, Title : Active views of passive cloud “regimes of regimes”,

2023 CloudSat/CALIPSO Annual Science Program Review, Title : Active views of passive cloud “regimes of regimes”

2022 AGU Fall meeting, Title: Examining the feasibility of detailed decomposition of radiative flux anomalies to cloud changes.

2021 AMS meeting, Title: Comparing the ISCCP and MODIS views of the planet’s cloudiness based on the mesoscale covariability of cloud altitude and extinction.

2021 AGU Fall meeting, Title: Combined analysis of coincident MODIS and ISCCP joint histograms for probing the differences in their respective cloud regime classifications.

2020 AGU Fall meeting, Title: Reconstructing subgrid cloud variability guided by CloudSat/CALIPSO observations.

2019 AeroCom/AeroSAT, Title: A Global view of Aerosol-Cloud interaction signals.

2019 AGU Fall meeting, Title: A new set of MODIS Cloud Regimes towards unification with ISCCP Weather States.
2018 MODIS/VIIRS Science Team Meeting, Title: Employing daily gridded MODIS data to study the impact of morning aerosol to afternoon clouds, precipitation, and radiation.
2018 AGU Fall meeting, Title: Apparent Relationships Between Morning Aerosol and Afternoon Cloud-Affected Quantities: Satellite and Model-Assisted Views.
2017 A-Train symposium, Title: Global hydrometeor vertical structure from CloudSat/CALIPSO: Radiative contributions, and insight about MODIS Cloud Regimes.
2017 Cloud Precipitation Center Seminar, Title: New insights about cloud Vertical Structure from CloudSat and CALIPSO observations.
2017 AGU Fall meeting, Title: Global hydrometeor vertical structure from CloudSat/CALIPSO: Radiative contributions, and insight about MODIS Cloud Regimes.
2016 AGU Fall meeting, Title: Cloud Vertical Structure variability within MODIS Cloud Regimes according to CloudSat-CALIPSO.
2015 MODIS/VIIRS Science Team Meeting, Title: Employing MODIS cloud regimes to find signals of aerosol-cloud-precipitation interactions.
2014 NASA Young Scientist forum, Title: A study of global MODIS cloud regimes.
2014 AGU Fall meeting, Title: Effects of aerosols on cloud and precipitation characteristics for distinct cloud regimes derived from MODIS.